

Introduction to Nutanix and AOS

DAY 3

Zookeeper

Ports: 9876

- Zookeeper is a centralized service for maintaining cluster information, naming, providing distributed synchronization and provide group services
- Zookeeper stores all cluster configuration including IPs, host information, host state and is based upon Apache Zookeeper
- This service runs on three nodes in the cluster, one of which is elected as a leader, in a RF2 cluster
- Leader receives all the requests and forwards it to its peers
- If a leader fails to respond, automatically a new leader is elected
- Zookeeper is accessed via an interface called Zeus.

Components of Zookeeper

Zeus

The interface used to access Zookeeper. You will sometimes see Zeus and Zookeeper used interchangeably

Zookeeper

The process used to store and maintain cluster configuration information

Command to gather Zookeeper configuration information: `zeus_config_printer`

Zookeeper Logs

Zookeeper.out : `/home/nutanix/data/logs/zookeeper.out`

Verbose log for Zookeeper. Contains detailed information on zeus/zookeeper communication between ZK nodes

Zookeeper Monitor : `/home/nutanix/data/logs/zookeeper_monitor.INFO|ERROR|FATAL`

Keeps track of zookeeper and other critical services. Helps when you want to confirm component health

Genesis.out : `/home/nutanix/data/logs/genesis.out`

This is the main log that will track zookeeper issues. Indicates if ZK nodes can communicate with each other and will note errors if any

Prism

Port: 9440

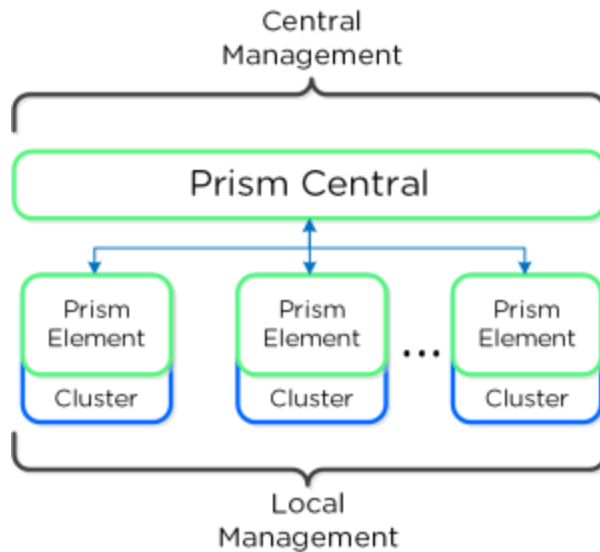
Prism is the process that provides distributed resource management platform for Nutanix environment.

Prism provides below key capabilities:

- **Interface:**
 - HTML5 (Prism Web Console)
 - REST API
 - NCLI
 - Powershell CMDlets
- **Management:**
 - Platform management, VM / Container CRUD
 - Policy definitions and compliance
 - Service configuration and status
 - Analytics and monitoring

Prism is broken down into two main components

Prism Central (PC)

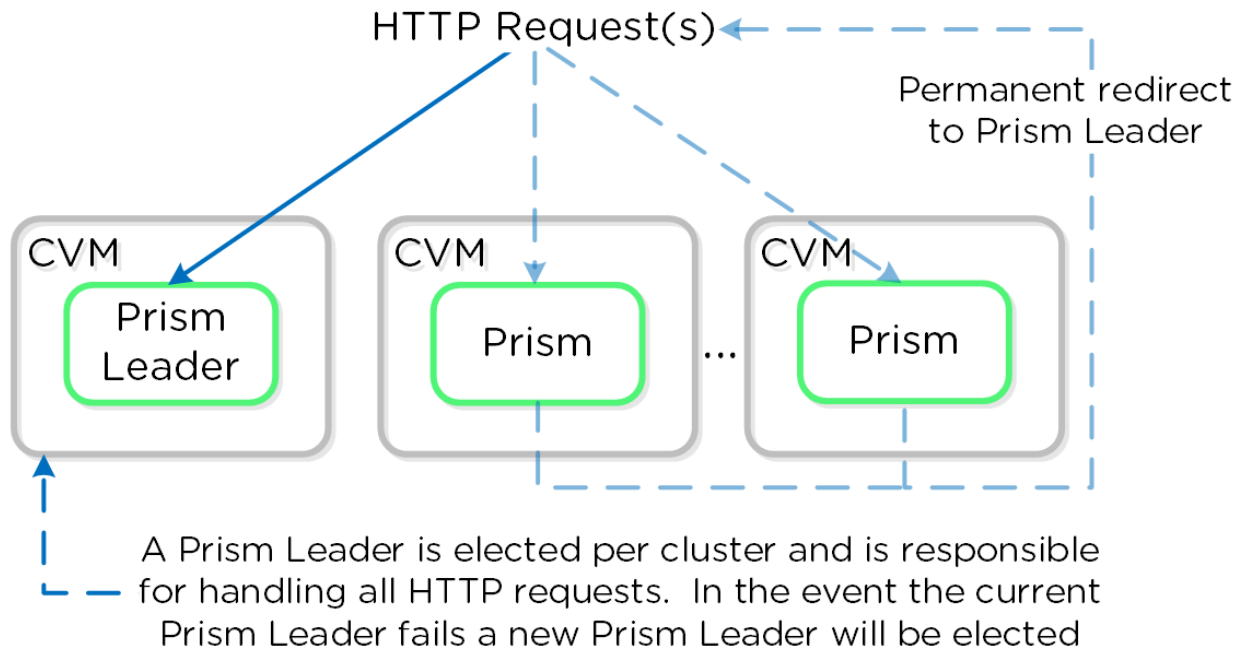


- Multi-cluster manager responsible for managing multiple Nutanix Clusters to provide a single, centralized management interface. Prism Central is an optional software appliance (VM) which can be deployed in addition to the AOS Cluster (can run on it).
- 1-to-many cluster manager

Prism Element (PE)

- Localized cluster manager responsible for local cluster management and operations. Every Nutanix Cluster has Prism Element built-in.
- 1-to-1 cluster manager

Prism Services:



- A Prism service runs on every CVM with an elected Prism Leader which is responsible for handling HTTP requests
- When a CVM which is not the Prism Leader gets a HTTP request it will redirect the request to the current Prism Leader

Prism listens on ports 80 and 9440. If HTTP traffic comes in on port 80, it is redirected to HTTPS on port 9440.

Cluster Virtual IP:

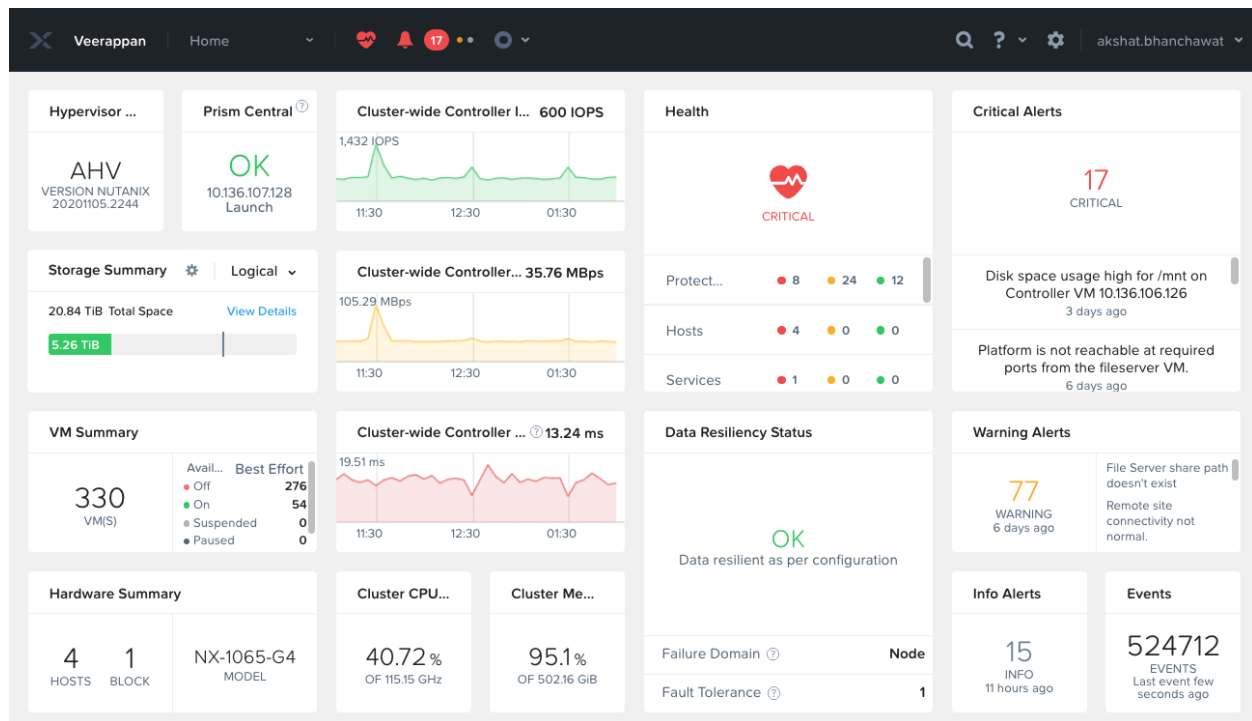
- Every cluster has a dedicated cluster external IP. This will be hosted by the current Prism Leader
- In the event of Prism Leader failure, cluster IP will be assumed by the newly elected Prism Leader

To determine Prism Leader: `curl localhost:2019/prism/leader`

Prism Element

The home page will provide detailed information on alerts, service status, capacity, performance, tasks, and much more. To get further information on any of them you can click on the item of interest.

Prism Element contains the following main pages:



- Home Page
 - Local cluster monitoring dashboard including detailed information on alerts, capacity, performance, health, tasks, etc. To get further information on any of them you can click on the item of interest.
- Health Page
 - Environment, hardware and managed object health and state information. Includes NCC health check status as well.
- VM Page
 - Full VM management, monitoring and CRUD (AOS)
- Storage Page
 - Container management, monitoring and CRUD
- Hardware
 - Server, disk and network management, monitoring and health. Includes cluster expansion as well as node and disk removal.

- Data Protection
 - DR, Cloud Connect and Metro Availability configuration. Management of PD objects, snapshots, replication and restore.
- Analysis
 - Detailed performance analysis for cluster and managed objects with event correlation
- Alerts
 - Local cluster and environment alerts

Core Data Path

Nutanix Distributed Storage Fabric is composed of the following high-level data structure:

- Storage Pool
- Container
- vDisk
- vBlock
- Extent
- Extent Group

Storage Pool

Key Role: Group of physical devices

- A storage pool is a group of physical storage devices including PCIe SSD, SSD, and HDD devices for the cluster
- The storage pool can span multiple Nutanix nodes and is expanded as the cluster scales
- In most configurations, only a single storage pool is leveraged

Container

Key Role: Group of VMs/files

- A container is a logical segmentation of the Storage Pool and contains a group of VM or files (vDisks)
- Some configuration options (e.g., RF) are configured at the container level, however are applied at the individual VM/file level
- Containers typically have a 1 to 1 mapping with a datastore (in the case of NFS/SMB)

vDisk

- vDisk is any file over 512KB on DSF including .vmdks and VM hard disks.
- vDisks are logically composed of vBlocks which make up the 'block map.'

| No artificial limits are imposed on the vdisk size on the DSF/stargate side

vBlock

Key Role: 1MB chunk of vDisk address space

- A vBlock is a 1MB chunk of virtual address space composing a vDisk
- For example, a vDisk of 100MB will have $100 \times 1\text{MB}$ vBlocks, vBlock 0 would be for 0-1MB, vBlock 1 would be from 1-2MB, and so forth
- These vBlocks map to extents which are stored as files on disk as extent groups

Extent

Key Role: Logically contiguous data

- An extent is a 1MB piece of logically contiguous data which consists of n number of contiguous blocks (varies depending on guest OS block size)
- Extents are written/read/modified on a sub-extent basis (aka slice) for granularity and efficiency. An extent's slice may be trimmed when moving into the cache depending on the amount of data being read/cached.

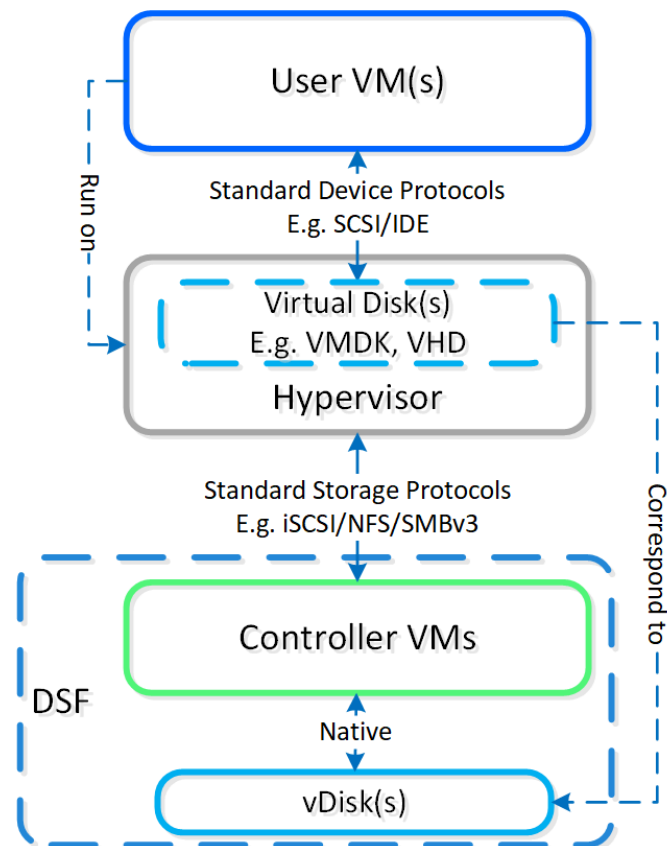
Extent Group

Key Role: Physically contiguous stored data

- An extent group is a 1MB or 4MB piece of physically contiguous stored data

- This data is stored as a file on the storage device owned by the CVM
- Extents are dynamically distributed among extent groups to provide data striping across nodes/disks to improve performance.
- Extent groups can now be either 1MB or 4MB depending on deduplication.

I/O Path and Cache

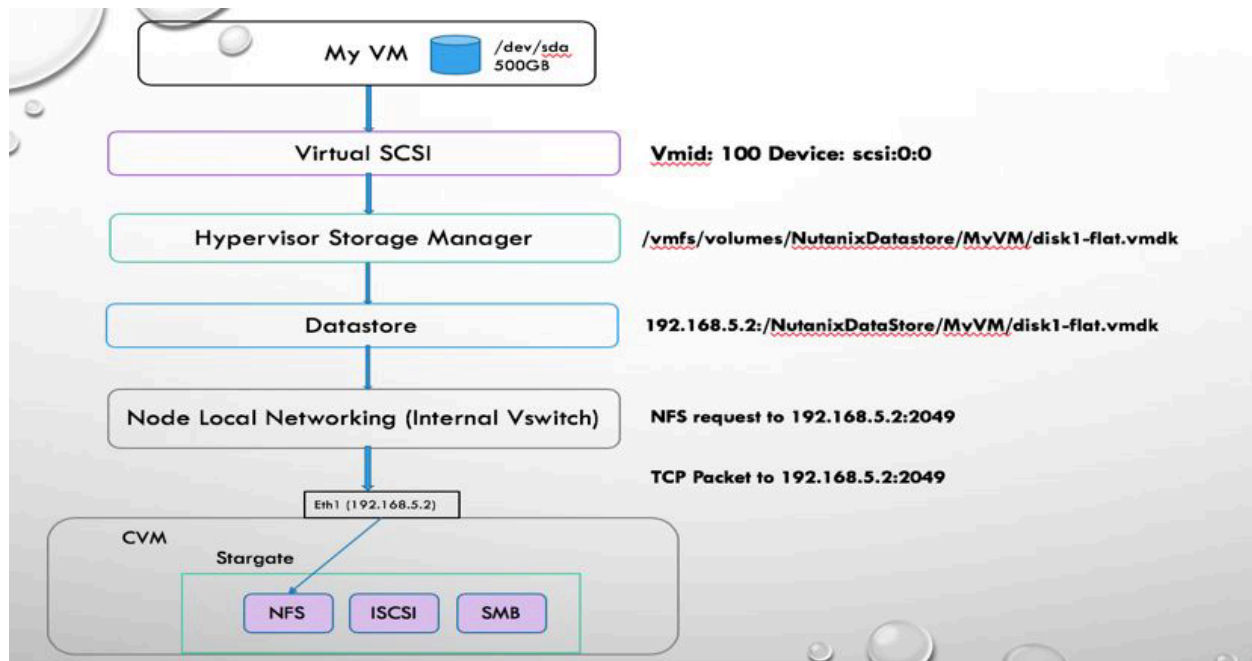


- **Guest OS (UVM) to virtual disk(s):** This remains unchanged with Nutanix. Depending on the hypervisor the guest OS will use a device driver to talk to a virtual disk device. Depending on the hypervisor this could be virtio-scsi (AHV), pv-scsi (ESXi), etc. The virtual disks will also vary based upon the hypervisor (e.g. vmdk, vhd, etc.)
- **Hypervisor to DSF (via CVM):** Communication between the hypervisor and Nutanix occurs via standard storage protocols (e.g. iSCSI, NFS, SMBv3) over the local interface of the CVM and hypervisor. At this point all communication

has been local to the host (there are scenarios where I/O will be remote (e.g. local CVM down, etc.)

- **Nutanix I/O path:** This is all transparent to the hypervisor and UVMs and it native to the Nutanix platform

Visualizing the I/O Path



Stargate

Ports used: 2009 (Inter-node and inter-component comm), 2049 (NFS), 3260 (iSCSI IET port), 3261 (iSCSI discovery), 3205 (iSCSI redirection), 445 (SMB adapter)

Stargate is responsible for all data management and I/O operations and is the main interface from the hypervisor (via NFS, iSCSI, or SMB). This service runs on every node in the cluster in order to serve localized I/O.

Stargate Page and Logs To access the Stargate page : [Links](http://<cvm_ip>:2009)
http://<cvm_ip>:2009

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|Resource|
|/qmodel_stats|Thread|
|/memory_stats|Memory|
|/estore_disk_stats|Extent Store Disk|
|/oplog_disk_stats|Oplog Disk|
|/cache_stats|Cache|
|Network|
|/rpc_stats|RPC|
|/conn_stats|RDMA Connections|
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|Queues|
|/admctl_queue_stats|Admctl Queue|
|/disk_queue_stats|Disk Queue|
|Extent Store|
|/disk_fitness_stats|Disk Fitness|
|/corrupt_egrups|Corrupt Egroups|
|DR|
|/replication_stats|Async Replication|
|/cg_controllers|Hosted CG Controllers|
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|Adapters and NFS|
|/iscsi|iSCSI|
|/smb|SMB|
|/nfs_pause_stats|NFS Pauses|
|VDisk|
|/vdisk_stats|VDisks|
|/oplog_flush_stats|Oplog Flush|
|/vdisk_hotspots|VDisk Hotspots|
|/hosted_vdisks|All Hosted VDisks|
|/vdisk_admctl|Admctl per-VDisk Queues|
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```

Stargate can have following categories of logs:

INFO / WARNING / ERROR /FATAL

Log location: `~/data/logs/stargate.out` and `~/data/logs/stargate.INFO`

CVM Autopathing (ha.py and iscsi_redirector)

Live discussion

Bringing down iscsi_redirector and stargate service on a node and showing the results.